

LAPORAN Praktikum 1 - Struktur Sistem Operasi



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**PROGRAM STUDI TEKNIK INFORMATIKA
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1. Operating-System Services Praktik

1.1: Menjelajahi Layanan OS

```
agen9@LDQ:~$ uname -a
Linux LDQ 6.6.87.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux
agen9@LDQ:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description: Ubuntu 24.04.3 LTS
Release: 24.04
Codename: noble
agen9@LDQ:~$ ps aux | head -20
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root             1  0.0  0.2 215404 12280 ?        Ss   22:01   0:00 /sbin/init
root             2  0.0  0.0   3072  1792 ?        Sl   22:01   0:00 /init
root             6  0.0  0.0   3104  1988 ?        Sl   22:01   0:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
root            42  0.0  0.2 422236 15688 ?        Ss   22:01   0:00 /usr/lib/systemd/systemd-journald
root            89  0.1  0.1 251336 6816 ?        Ss   22:01   0:00 /usr/lib/systemd/systemd-udevd
systemd+       104  0.0  0.2 21456 12832 ?        Ss   22:01   0:00 /usr/lib/systemd/systemd-resolved
systemd+       165  0.0  0.1 91024 7552 ?        Ssl  22:01   0:00 /usr/lib/systemd/systemd-timesyncd
root           165  0.0  0.0   4236  2432 ?        Ss   22:01   0:00 /usr/sbin/cron -f -P
message+       166  0.0  0.0   9516  4992 ?        Ss   22:01   0:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --sysl
og-only
root           173  0.0  0.1 17964 8320 ?        Ss   22:01   0:00 /usr/lib/systemd/systemd-logind
root           175  0.0  0.2 175504 11904 ?       Ssl  22:01   0:00 /usr/libexec/ssl-proc-service -vv
root           177  0.0  0.0   3160  1920 hvcB    Ss+  22:01   0:00 /sbin/agetty -o -p -- \u --nolinux --keep-baud - 115200,38400,9600 vt220
syslog         193  0.0  0.0 222588 5584 ?        Ssl  22:01   0:00 /usr/sbin/rsyslogd -n -lNONE
root           197  0.0  0.0   3116  1664 tty1    Ss+  22:01   0:00 /sbin/agetty -o -p -- \u --nolinux - linux
root           203  0.0  0.3 107832 22528 ?       Ssl  22:01   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-si
gnal
root           403  0.0  0.0   3076   768 ?        Ss   22:02   0:00 /init
root           404  0.0  0.0   3092  1152 ?        S   22:02   0:00 /init
agen9          408  0.0  0.0   6972  5248 pts/0    Ss   22:02   0:00 -bash
root           497  0.0  0.0   6824  4480 pts/1    Ss   22:02   0:00 /bin/login -f
agen9@LDQ:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           5.6Gi        472Mi       4.8Gi        3.5Mi       510Mi       5.2Gi
Swap:           2.0Gi          0B         2.0Gi
agen9@LDQ:~$ lscpu
Architecture: x86_64
CPU op-mode(s): 32-bit, 64-bit
Address sizes: 39 bits physical, 48 bits virt
Byte Order: Little Endian
CPU(s): 12
On-line CPU(s) list: 0-11
Vendor ID: GenuineIntel
```

```
agen9@LDQ:~$ cat /proc/cpuinfo
Vendor ID: GenuineIntel
Model name: 12th Gen Intel(R) Core(TM) i5-12450HX
CPU family: 6
Model: 151
Thread(s) per core: 2
Core(s) per socket: 6
Socket(s): 1
Stepping: 2
BogoMIPS: 5376.00
Flags: fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtscp lm constant_tsc rep_good nopl xtopology tsc_reliable nontstop tsc_cquid tsc_knom _freq pni pclmulqdq vmx ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdprnd hypervisor lahf_lm abm 3dnowprefetch ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow opt_vpid ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid rdpseed adx smap clflushopt t clwb sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni vnni1 uap waitpkg gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrm mml_clear serialize_flush_lld arch_capabilities

Virtualization features:
Virtualization: VT-x
Hypervisor vendor: Microsoft
Virtualization type: full
Caches (sum of all):
L1d: 288 KiB (6 instances)
L1i: 192 KiB (6 instances)
L2: 7.5 MiB (6 instances)
L3: 12 MiB (1 instance)
NUMA:
NUMA node(s): 1
```

```
agen9@LDQ:~$ cat /proc/meminfo
MemTotal: 5888000 kB
MemFree: 5088000 kB
MemAvailable: 5248000 kB
Buffers: 102400 kB
Cached: 102400 kB
SwapCached: 0 kB
Active: 102400 kB
Inactive: 102400 kB
Active(anon): 102400 kB
Inactive(anon): 102400 kB
Active(file): 102400 kB
Inactive(file): 102400 kB
Unreclaimable: 102400 kB
Mlocked: 102400 kB
SwapTotal: 2048000 kB
SwapFree: 2048000 kB
Dirty: 102400 kB
Writeback: 102400 kB
CommitLimit: 1024000 kB
Committed_AS: 102400 kB
PageTables: 102400 kB
SReclaimable: 102400 kB
SUnreclaimable: 102400 kB
KernelStack: 102400 kB
Page_Remediation: 102400 kB
VmallocTotal: 1024000 kB
VmallocUsed: 102400 kB
VmallocChunk: 102400 kB
Hardware_Numa: 1
HugePages_Total: 0
HugePages_Free: 0
HugePages_Rsvd: 0
HugePages_Surp: 0
Hugepagesize: 2048 kB
DirectMap4k: 102400 kB
DirectMap2M: 102400 kB
DirectMap1G: 102400 kB
MmapStaleData: 102400 kB
RegFileDataSampling: Mitigation: Clear Register File
Retbleed: Mitigation: Enhanced IBRS
Spec rstack overflow: Not affected
Spec store bypass: Mitigation: Speculative Store Bypass disabled via prctl
Spectre v1: Mitigation: usercopy/swapgs barriers and __user pointer sanitization
Spectre v2: Mitigation: Enhanced / Automatic IBRS, IBPB conditional, RSB filling; PBRSSB-eIBRS SW sequence; BMI BMI_Dis_
Srbds: Not affected
Taxi anyne abort: Not affected
agen9@LDQ:~$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda 8:0 0 388.4M 1 disk
sdb 8:16 0 186M 1 disk
sdc 8:32 0 8G 0 disk [SWAP]
sdd 8:48 0 1T 0 disk /mnt/wslg/distro
```

Buat file `os_services.c`:

2. User and Operating-System Interface Praktik

[illegible]

```

$ cd /tmp
$ echo "Hello World" > test_dir/test_dir
$ ls -la /tmp/test_dir
total 4
drwxr-xr-x 2 root root 4096 Oct 18 22:18 test_dir
-rw-r--r-- 1 root root   12 Oct 18 22:18 test
$ cat /tmp/test_dir/test_dir
Hello World
$ cat /tmp/test_dir/test
Hello World
$ sleep 30 &
[1] 30
[2] 30
[3] 30
[4] 30
[5] 30
[6] 30
[7] 30
[8] 30
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```

Buat file `simple_shell.c`:

3. System Calls

Buat file `process_syscalls.c`:

Buat file file_syscalls.c:

Buat file `info_syscalls.c`:

[illegible]

4.1: Menggunakan system

Praktik 4.2: Membuat Custom System Service

```

ngmng@LDQ: ~$
[snaps] password for ngmng:
[snaps]~$ sudo systemctl start my_daemon
[snaps]~$ sudo systemctl status my_daemon
Failed to start my_daemon.service: Unit my_daemon.service not found.
ngmng@LDQ: ~$ sudo nano /etc/systemd/system/my_daemon.service
[snaps]~$ sudo nano /etc/systemd/system/my_daemon.service
[snaps]~$ sudo nano /etc/systemd/system/my_daemon.service
[snaps]~$ sudo nano /etc/systemd/system/my_daemon.service
[snaps]~$ sudo systemctl daemon-reload
[snaps]~$ sudo systemctl start my_daemon
[snaps]~$ sudo systemctl status my_daemon
Unit my_daemon.service failed because the control process exited with error code.
See "systemctl status my_daemon.service" and "journalctl -u my_daemon.service" for details.
[snaps]~$ sudo systemctl status my_daemon
Unit my_daemon.service - My Custom Daemon
Loaded: loaded (/etc/systemd/system/my_daemon.service; disabled; preset=enabled)
Active: failed (Result: exit-code) since Mon 2023-10-30 18:17:08 GMT; 2min 28s ago
Process: 2876 ExecStart=/path/to/my_daemon (code=exited, status=255/CU
CPU: 2ms
Oct 30 18:17:08 LDQ systemd[1]: my_daemon.service: Scheduled restart job,
Oct 30 18:17:08 LDQ systemd[1]: my_daemon.service: Start request repeated
Oct 30 18:17:08 LDQ systemd[1]: my_daemon.service: Failed with result 'fail
Oct 30 18:17:08 LDQ systemd[1]: Failed to start my_daemon.service - My Cus
[snaps]~$ sudo systemctl status my_daemon.service - My Custom Daemon
Unit my_daemon.service - My Custom Daemon
Loaded: loaded (/etc/systemd/system/my_daemon.service; disabled; preset=
Active: failed (Result: exit-code) since Mon 2023-10-30 18:17:08 GMT; 2min
Process: 2876 ExecStart=/path/to/my_daemon (code=exited, status=255/CU
CPU: 2ms
Oct 30 18:17:08 LDQ systemd[1]: my_daemon.service: Scheduled restart job,
Oct 30 18:17:08 LDQ systemd[1]: my_daemon.service: Start request repeated
Oct 30 18:17:08 LDQ systemd[1]: my_daemon.service: Failed with result 'fail
Oct 30 18:17:08 LDQ systemd[1]: Failed to start my_daemon.service - My Cus

```


Buat file `os_specific.c`:

[illegible]

7.1: Kernel Module Sederhana

[illegible][illegible]

Praktik 8.2: Microkernel vs Monolithic

[illegible]

Praktik 8.3: Program - Virtual File System

Buat file `procfs_read.c`:

```

root@kali:~# nano procfs_read.c
procfs_read.c:5 gcc -o procfs_read procfs_read.c
procfs_read.c:5 function 'main'
procfs_read.c:22:46: error: implicit declaration of function 'getpid' is invalid [-Wimplicit-function-declaration]
22 | return(proc_stat, sizeof(proc_stat), "/proc/0/stat", "pid(0)");
   |                                     ^~~~~~
root@kali:~# ./procfs_read
=== CPU Information ===
processor      : 0
processor_id    : GenuineIntel
cpu family     : 6
model          : 231
model name     : Intel(R) Core(TM) i5-12400H
stepping       : 2
microcode     : 0x00000000
cpu mhz        : 2800.002
cache size    : 27360 KB
physical id    : 0

=== Memory Information ===
memTotal      : 30176MB MB
memFree       : 25280MB MB
memAvailable  : 93627MB MB
buffers       : 6716 MB
cached        : 22762MB MB

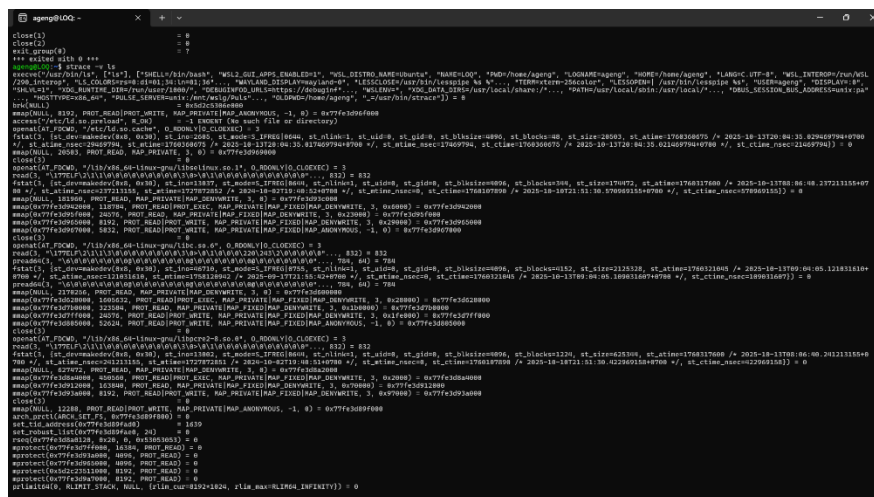
=== Process Information ===
name          : procfs_read
uid           : 0002
state         : R (running)
pid           : 0
ppid          : 3155
uid           : 0
gid           : 0
ppid          : 3157
memFree       : 0
uid           : 1000
gid           : 1000
ppid          : 1000

```

9. Building and Booting an Operating System

Praktik 9.1: Analisis Boot Process

[illegible]



Praktik 10.2: Using ltrace

[illegible]

Buat program dengan bug buggy.c:

Praktik 10.4: Kernel Debugging dengan dmesg

[illegible]

[illegible][illegible]

Praktik 10.5: Memory Leak Detection dengan Valgrind

[illegible]

Praktik 10.6: Performance Analysis

```
agen@WLO: ~  
$ gcc -g -o memleak memleak.c  
memleak.c: In function 'main':  
memleak.c:2:19: warning: implicit declaration of function 'malloc' [-Wimplicit-function-declaration]  
2 |   int *ptr = (int*)malloc(100 * sizeof(int));  
   |                   ^~~~~~  
memleak.c:1:1: note: include '<stdlib.h>' or provide a declaration of 'malloc'  
1 | #include <stdlib.h>  
   | ^~~~~~  
memleak.c:2:19: note: incompatible implicit declaration of built-in function 'malloc' [-Wbuiltin-declaration-mismatch]  
2 |   int *ptr = (int*)malloc(100 * sizeof(int));  
   |                   ^~~~~~  
memleak.c:2:19: note: include '<stdlib.h>' or provide a declaration of 'malloc'  
agen@WLO: ~$ valgrind --leak-check=full ./memleak  
==1765== Memcheck, a memory error detector  
==1765== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.  
==1765== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info  
==1765== Command: ./memleak  
==1765==  
==1765== HEAP SUMMARY:  
==1765==    in use at exit: 400 bytes in 1 blocks  
==1765== total heap usage: 1 allocs, 0 frees, 400 bytes allocated  
==1765==  
==1765== 400 bytes in 1 blocks are definitely lost in loss record 1 of 1  
==1765==    at 0x48502F8: malloc (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)  
==1765==    by 0x109156: main (memleak.c:2)  
==1765==  
==1765== LEAK SUMMARY:  
==1765==    definitely lost: 400 bytes in 1 blocks  
==1765==    indirectly lost: 0 bytes in 0 blocks  
==1765==    possibly lost: 0 bytes in 0 blocks  
==1765==    still reachable: 0 bytes in 0 blocks  
==1765==    suppressed: 0 bytes in 0 blocks  
==1765==  
==1765== For lists of detected and suppressed errors, rerun with: -s  
==1765== ERROR SUMMARY: 1 errors from 1 contexts (suppressed: 0 from 0)  
agen@WLO: ~$ sudo apt install linux-tools-common linux-tools-generic  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
linux-tools-common is already the newest version (6.8-83.83).  
linux-tools-generic is already the newest version (6.8-83.83).  
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.  
agen@WLO: ~$ sudo perf trace ls  
WARNING: perf not found for kernel 6.6.87.2-microsoft  
  
You may need to install the following packages for this specific kernel:  
  linux-tools-6.6.87.2-microsoft-standard-WSL2  
  linux-cloud-tools-6.6.87.2-microsoft-standard-WSL2  
  
You may also want to install one of the following packages to keep up to date:  
  linux-tools-standard-WSL2  
  linux-cloud-tools-standard-WSL2  
agen@WLO: ~$ sudo perf top  
WARNING: perf not found for kernel 6.6.87.2-microsoft  
  
You may need to install the following packages for this specific kernel:  
  linux-tools-6.6.87.2-microsoft-standard-WSL2
```